8:08 vulms.vu.edu.pk/Qui: 25

Question # 10 of 10 (Start time: 08:08:12 PM, 21 August 2022)

The value of $\int_1^{\infty} \frac{dx}{x^2}$ _____.

Select the correct option

○	1
○	4
○	0
○	3

Time L

l Geometry (Quiz 2)

Quiz Start Time: 07:56 PM, 21 A

:06:42 PM, 21 August 2022)

To

The region bounded by the curves $y = \sqrt{x}$, $y = 1$ *and* $x = 4$

Reload Math

$\frac{1}{2}$

$\frac{5}{3}$

None of these

0

Click to Save Answer & Move to Next Q

Find the area of the region between the $x-axis$, the $f(x)=x^3-x^2-2x;\ -1\leqslant x\leqslant 2$

rrect option

Reload Math E

$\frac{3}{2}$

$\frac{37}{12}$

$\frac{45}{4}$

None of these

Click to Save Answer & Move to Next Qu

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MTH101 - Calculus And Analytical Geometry (Quiz 2) Quiz Start Time: 08:14 PM, 21 Aug

Question # 3 of 10 (Start time: 08:16:34 PM, 21 August 2022) Total

If we change the letter for the variable of integration but don't change the limits, then the values of the definite integral are unchanged.

Select the correct option

- False
- True

Click to Save Answer & Move to Next Que

8:15 46%
vulms.vu.edu.pk/Quiz/ 26

BC220205096: HUSSNAIN ZULFIQAR

MTH101 - Calculus And Analytical Geometry (Quiz 2) Quiz Start Time

Question # 2 of 10 (Start time: 08:15:43 PM, 21 August 2022)

First fundamental theorem of calculus tells us how to evaluate the ..... in a quick way.

Select the correct option

- Definite integral
- None of these.
- Differential
- Indefinite integral

Click to Save Answer &

8:14 46%
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26

Question # 1 of 10 (Start time: 08:14:50 PM, 21 August 2022)
Total M

The volume of a cylinder is the area of a cross section of the cylinder multiplied by the _____ of the cylinder.

Select the correct option

- Diameter
- Height
- Radius
- Base

Click to Save Answer & Move to Next Questi

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○
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BC220205096: HUSSNAIN ZULFIQAR

Time

MTH101 - Calculus And Analytical Geometry (Quiz 2)

Quiz Start Time: 08:14 PM, 21

Question # 4 of 10 (Start time: 08:17:25 PM, 21 August 2022)

EVERY continuous function on an interval has an anti-derivative .....

Select the correct option

- On that interval
- outside of interval

Click to Save Answer & Move to Next

8:18 45%

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BC220205096: HUSSNAIN ZULFIQAR

Time Left

MTH101 - Calculus And Analytical Geometry (Quiz 2)

Quiz Start Time: 08:14 PM, 21 Augu

Question # 5 of 10 (Start time: 08:18:06 PM, 21 August 2022)

Total

For the adjacent intervals, [a,c] and [c,b],where c is any number,

$$\int_a^b f(x)dx =$$

Select the correct option

Reload Math Equ

- $\int_a^c f(x)dx + \int_b^a f(x)dx$
- $\int_a^c f(x)dx + \int_c^b f(x)dx$
- None of these
- $\int_a^b f(x)dx + \int_c^a f(x)dx$

Click to Save Answer & Move to Next Ques

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BC220205096: HUSSNAIN ZULFIQAR

MTH101 - Calculus And Analytical Geometry (Quiz 2) Quiz Start Time: 08:

Question # 6 of 10 (Start time: 08:19:09 PM, 21 August 2022)

The value of $\int_0^{\frac{\pi}{4}} \tan^2 x \, dx$ _____.

Select the correct option

Rel

- 0
- None of the above
- $1 - \frac{\pi}{4}$
- $\frac{\pi}{4} - 1$

Click to Save Answer & Mov

8:20 PM

Quiz
vulms.vu.edu.pk

BC220206409: AMAIMA RAZZAQ

MTH101 - Calculus And Analytical Geometry (Quiz 2) **Quiz Start Time:**

Question # 2 of 10 (Start time: 08:20:27 PM, 21 August 2022)

The integral of f(x)=sin(2x) from x=0 to x=pi is .....

Select the correct option

- 0
- 2
- 1
- None of thes.

Click to Save Answer &

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26

MTH101 - Calculus And Analytical Geometry (Quiz 2) Quiz Start Time

Question # 9 of 10 (Start time: 08:21:32 PM, 21 August 2022)

The value of $\int_1^x y^2 dy =$ _____.

Select the correct option

- $\frac{y^3}{3} - \frac{1}{3}$
- $\frac{x^3}{3} - \frac{1}{3}$

Click to Save Answer &

8:19 45%
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26

MTH101 - Calculus And Analytical Geometry (Quiz 2) Quiz Start Time: 0

Question # 7 of 10 (Start time: 08:19:51 PM, 21 August 2022)

The value of $\int_{1}^{10} 3x^2\,dx$ _____.

Select the correct option

- 999
- 333
- 99
- 33

Click to Save Answer & Mo

8:22 45%

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Time Left 88 sec(s)

MTH101 - Calculus And Analytical Geometry (Quiz 2)

Quiz Start Time: 08:14 PM, 21 August 2022

Question # 10 of 10 (Start time: 08:22:44 PM, 21 August 2022)

Total Marks: 1

The conclusion of the 2nd fundamental theorem of calculus can be expressed in words as: "the derivative of an integral of a function is that....."

Select the correct option

- none of these
- Next function
- Previous function
- Original function

Click to Save Answer & Move to Next Question

III O <

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Question # 6 of 10 (Start time: 08:26:06 PM, 21 August 2022)

If the upper and lower limits for the definite integral are the same, then $\int_a^a f(x)dx =$

Select the correct option

Reload

- positive integer
- none of these
- negative integer
- zero

Click to Save Answer & Move

8:30 PM

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Question # 9 of 10 (Start time: 08:30:37 PM, 21 August 2022)

F(x) (antiderivative of f) was determined to be 4x+C where C denotes the .....

Select the correct option

- Integration constant.
- Differentiation constant.

Click to Save Answer & Mo

Question # 9 of 10 (Start time: 08:55:07 PM, 21 August 2022)

The value of $\int_0^x t^3\,dt =$ _____.

Select the correct option

○	$\frac{t^4}{4}$
○	$\frac{x^4}{4}$
○	$\frac{t^4}{4} - 1$
○	None of the above

The volume of solid obtained when the region under the curve $y = x^3$ over the interval [1,3] is revolved about the x-axis is

Select the correct option

○	$V = \int_1^3 \pi x^3 \, dx$
○	$V = \int_1^3 \pi x^6 \, dx$
○	$V = \int_1^3 x^6 \, dx$
○	$V = \int_1^3 \pi x^3 \, dy$

8:32 PM

Quiz
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Question # 10 of 10 (Start time: 08:31:58 PM, 21 August 2022)

In integration of $f(x) = x(x^2+1)^3$ from x=0 to x=2 by substitution method, we take $u = x^2 + 1$ then du= .....

Select the correct option

Reload Mc

○	dx
○	$2xdx$
○	1
○	xdx

Click to Save Answer & Move to Ne

Question # 6 of 10 (Start time: 08:52:12 PM, 21 August 2022) T

If the solid is revolved around the y-axis and generates a solid with a circular cross section of radius g(y) at y. Then the area of this cross section is

Select the correct option

Reload Math

○	$\pi[g(y)]^2$
○	$\pi\,[g(y)]$
○	$\pi r[g(y)]^3$
○	$[g(y)]^2$

Question # 7 of 10 (Start time: 08:53:08 PM, 21 August 2022)

The volume of solid obtained when the region under the curve x=y over the interval [1,4] is revolved about the y-axis is

Select the correct option

○	$V = \int_1^4 \pi dx$
○	$V = \int_1^4 \pi y dy$
○	$V = \int_1^4 y^2 dy$
○	$V = \int_1^4 \pi y^2 dy$

Question # 8 of 10 (Start time: 08:54:01 PM, 21 August 2022)

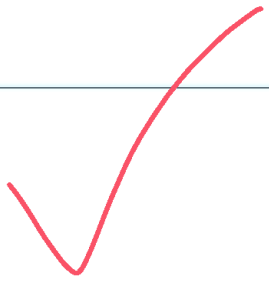
If the function and limits of definite integral are the same and variable of integration are changed, i.e

$$\int_a^b f(x)dx = \int_a^b f(t)dt$$

Then the answer would be

Select the correct option

○	changed
○	do not changed



Question # 3 of 10 (Start time: 08:48:11 PM, 21 August 2022)

If

$$\int_0^2 (x^2+1)dx = \frac{14}{3}$$

Then the solution of

$$\int_2^0 (x^2+1)dx =$$

will be....

Select the correct option

○	14/3
○	-14/3
○	-3/14
○	none of these

Question # 4 of 10 (Start time: 08:49:56 PM, 21 August 2022)

If sinx=t then dt=.....

Select the correct option

○	sinxdt
○	cosxdx
○	-cosxdx
○	sinxdx

Question # 5 of 10 (Start time: 08:50:56 PM, 21 August 2022)

The integral of f(x) =cos(2x) from x=0 to x=pi is .....

Select the correct option

○	3(pi)
○	None of these.
○	0
○	2(pi)

Which of the following is the definite integral of f(x) =x^2 from x=1 to x=2

Select the correct option

○	8/3
○	7/3
○	None of these.
○	7

Question # 1 of 10 (Start time: 08:46:40 PM, 21 August 2022)

If f is continuous at every point of $[a, b]$ and F is anti - derivative of f on $[a, b]$, then

Select the correct option

○	none of these
○	$\int_a^b f(x)dx = F(a) + F(b)$
○	$\int_a^b f(x)dx = F(a) - F(b)$
○	$\int_a^b f(x)dx = F(b) - F(a)$ ✓

Click to Sa

8:02 vulms.vu.edu.pk/Qui: 25

MTH101 - Calculus And Analytical Geometry (Quiz 2) Quiz Start Time: 07:5

Question # 5 of 10 (Start time: 08:02:45 PM, 21 August 2022)

$\int_a^b f(x)dx = _____.$

Select the correct option

Rela

- $-\int_b^a f(x)dx$
- $\int_b^a f(x)dx$

Click to Save Answer & Move

8:03 vulms.vu.edu.pk/Qui: 25

Question # 6 of 10 (Start time: 08:03:46 PM, 21 August 2022)

..... can be used to find the area under the curve and volume of the solids.

Select the correct option

○	Limit
○	Integration
○	none of these.
○	Differentiation

8:04 vulms.vu.edu.pk/Qui: 25

Which of the following is the definite integral of f(x) =x^2 from x=1 to x=2

Select the correct option

○	None of these.
○	7
○	8/3
○	7/3

7:58
vulms.vu.edu.pk/Qui
25

Question # 2 of 10 (Start time: 07:58:08 PM, 21 August 2022)

The integral of f(x) =sin(x+1) from x=0 to x=1 is .....

Select the correct option

○	sin(1)-sin(2)
○	cos(2)-cos(1)
○	cos(2)-1
○	cos(1)-cos(2)

7:56

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25

Time Left 64 sec(s)

metry (Quiz 2)

Quiz Start Time: 07:56 PM, 21 August 2022

PM, 21 August 2022)

Total Marks: 1

The area of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

Reload Math Equations

None of these

$\frac{1}{4}\pi(a^2 + b^2)$

πab

$\pi(a + b)$

Click to Save Answer & Move to Next Question

alytical Geometry (Quiz 2) Quiz Start Time:

me: 07:59:19 PM, 21 August 2022)

$The\ bounded\ region\ between\ the\ parabola\ y = 4x^2\ and\ the\ line\ y = 6x - 2$

$None\ of\ these$

$\frac{1}{3}$

$\frac{1}{12}$

$\frac{1}{6}$

Click to Save Answer &

8:20 45%
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26

BC220205096: HUSSNAIN ZULFIQAR

MTH101 - Calculus And Analytical Geometry (Quiz 2) Quiz Start Time: 0

Question # 8 of 10 (Start time: 08:20:32 PM, 21 August 2022)

The value of the $\int_0^1 t^3 dt =$ _____.

Select the correct option

- 1/3
- 2/3
- 4/3
- 1/4

Click to Save Answer & M

19:26 4G

$$Evaluate \frac{d}{dx}\int_1^x t^2 dt =$$

$-x^2$

x^2

$3x^2$

Click to Sav

19:27

Quiz
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H101 - Calculus And Analytical Geometry (Quiz 2)

ıstion # 9 of 10 (Start time: 07:27:22 PM, 21 August 2022)

' the function and limits of definite integral are the same and variable of integration are changed, i.e

$$\int_a^b f(x)dx = \int_a^b f(t)dt$$

hen the answer would be

ect the correct option

- do not changed
- changed

Click t

19:29 Quiz vulms.vu.edu.pk

$\int_a^b f(x)dx =$ _____.

Select the correct option

- $\int_b^a f(x)dx$
- $-\int_b^a f(x)dx$

19:25

Quiz
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Question # 7 of 10 (Start time: 07:25:11 PM, 21 August 2022)

We cannot evaluate definite integrals by substitution method.

Select the correct option

- True
- False



19:23

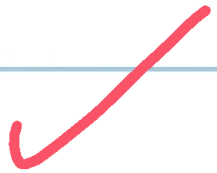
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uestion # 6 of 10 (Start time: 07:23:38 PM, 21 August 2022)

The volume V of a cylinder with base area A and height h is calculated by -----

elect the correct option

- V = A h^3
- V = A h^2
- V = 2A h
- V = A h



19:22

Quiz
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The value of $\int_0^{\frac{\pi}{4}} \tan^2 x \, dx$ _____.

Select the correct option

○	$1-\frac{\pi}{4}$
○	0
○	$\frac{\pi}{4}-1$
○	None of the above

Click to Save An

19:18

Quiz
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Question # 2 of 10 (Start time: 07:18:16 PM, 21 August 2022)

..... is used to prove the first fundamental theorem of calculus.

Select the correct option

- Intermediate value theorem
- None of these.
- Mean value theorem for the derivatives
- Mean value theorem for the integrals

19:19

Quiz
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07:19:17 PM, 21 August 2022)

The bounded region between the parabola $y = 4x^2$ *and the line* $y = 6x - 2$

Relo

$\frac{1}{6}$

None of these

$\frac{1}{3}$

$\frac{1}{12}$

Click to Save Answer & Move

19:21

BC220204480: AYESHA MUSHTAQ

Time Left 74 sec(s)

MTH101 - Calculus And Analytical Geometry (Quiz 2)

Quiz Start Time: 07:16 PM, 21 August 2022

Question # 4 of 10 (Start time: 07:20:48 PM, 21 August 2022)

Total Marks: 1

If the solid is revolved around the y-axis and generates a solid with a circular cross section of radius g(y) at y. Then the area of this cross section is

Select the correct option

Reload Math Equations

- $[g(y)]^3$
- $\pi[g(y)]^2$
- $\pi+[g(y)]^3$
- $\pi[g(y)]$

Click to Save Answer & Move to Next Question

19:17

Quiz
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Question # 1 of 10 (Start time: 07:16:34 PM, 21 August 2022)

The value of $\int_1^x y^2 dy =$ _____.

Select the correct option

○	$\frac{y^3}{3} - \frac{1}{3}$
○	$\frac{x^3}{3} - \frac{1}{3}$ ✓

19:16

Quiz
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BC220204480: AYESHA MUSHTAQ

Time Left 87 sec(s)

MTH101 - Calculus And Analytical Geometry (Quiz 2)

Quiz Start Time: 07:16 PM, 21 August 2022

Question # 1 of 10 (Start time: 07:16:34 PM, 21 August 2022)

Total Marks: 1

The value of $\int_{1}^{2} y^2 dy =$ .....

Select the correct option:

Reload Math Equations

- $\frac{7}{3} - \frac{1}{3}$
- $\frac{7}{3} - \frac{1}{3}$

Click to Save Answer & Move to Next Question